WACE Mathematics Methods · Cheat Sheet (Patterns + Notes)

General rules: show that: write every step · for optimisation questions that require using calculus, you must verify the extremum (f″ or sign table, 1 mark) · calculator in radians (if π/180 appears when differentiating sin, it is set to degrees) · include units in application questions · in explanations, state clearly "which quantity + when + how much it changes + units", in context · probabilities to at least 4 decimal places (if the question specifies 3 decimal places, exact or integer, follow that) · for CA questions, write the equation you solve and the key output; do not write calculator syntax · read the question carefully for "first / initially" (t = 0), "more than" (≥ k+1), "at least"

1 Differentiation and its applications

① Direct differentiation (CF opening 2–4 marks)

- First decide the structure: product → product; fraction → quotient; composite → chain, often used together (e.g. $x^2 \ln(4x + 3)$).
- Basic formulas: $(e^{kx})' = ke^{kx}$; $(\ln f)' = f'/f$; $(\sin kx)' = k\cos kx$; $(\tan x)' = 1/\cos^2 x$ (2017A-Q10 requires proof using the quotient rule).
- ⚠ The method mark requires writing the two-term structure and at least one term correct; for $(2x^3+1)^5$ do not forget to multiply by $6x^2$; when differentiating e^{-x} do not miss the negative sign; if it says "do not simplify", you do not need to simplify.

② Differentiation from table / graph values

- First write the general formula (1 mark): $(fg)' = f'g + fg'$; $(g/h)' = (g'h - gh')/h^2$; $[h(g(x))]' = h'(g(x))\cdot g'(x)$.
- ⚠ For composites, evaluate the inner layer first: $h'(g(1)) = h'(3)$, not $h'(1)$ (2019 report: poor responses on composite differentiation).
- Graph: the gradient of a straight-line segment is the derivative; at a corner (sharp point) it is not differentiable.

③ "rate of change of f′(x)" / interpreting the derivative

- This means finding f″(a) (recognising this is worth 1 mark by itself).
- Explanation template: "When x = a, the gradient of f is increasing at a rate of … per unit of x"; $f'(a) = 0$ and $f''(a) < 0 \Rightarrow$ local maximum.

④ Full function analysis (10–15 mark extended question)

- Factorise f′, state $e^x > 0$ (never zero) → solve $f' = 0$ → substitute back into the original function to find y-coordinates (marked separately).
- Nature: $f''(a) > 0$ minimum, < 0 maximum; ⚠ when $f''(a) = 0$ you must use a sign table instead (2023 report).
- Points of inflection: $f'' = 0$ and f″ changes sign on both sides. ⚠ $f' = f'' = 0$ alone is not enough (x^4 has a minimum at 0).
- Sketching: label coordinates of intercepts, stationary points and points of inflection; write equations of asymptotes (for $\ln(x + 3)$, domain $x > -3$ and vertical asymptote are worth 1 mark separately); use open/closed circles at endpoints; concavity matches the sign of f″.
- Example: 2021A-Q12 total 15 marks = differentiation 2 + stationary points and nature 7 + points of inflection 2 + sketching 4.

⑤ Finding coefficients from conditions / sketching from conditions

- Passes through point $f(a) = b$; turning point $f'(a) = 0$; point of inflection $f''(a) = 0$; area $\int f = A$. First write the general forms of f′ and f″.
- Example: 2018A-Q15's "growth pauses at t = 6, then continues increasing" ⇒ horizontal point of inflection ⇒ $P'(6) = 0$ and $P''(6) = 0$.
- Sketching from conditions (2022F-Q5): $f''(-1) > 0 \Rightarrow$ minimum; only two stationary points and $f'(2) > 0 \Rightarrow x = 1$ is a horizontal point of inflection.

⑥ Optimisation ★

- show that: use the constraint (fixed volume, perimeter 500 m, Pythagoras relationship) to eliminate one variable, substitute into the objective function, and simplify to the form given in the question.
- Set derivative = 0, reject unreasonable solutions (r = 0, θ outside range) → verify: calculate $A''(x_0) < 0$ at the stationary point, or draw a sign table → answer with all dimensions (both radius and height), round as required and include units.
- If the question says "You do not need to verify", you can omit verification; if only integers are allowed, compare the values of the two adjacent integers (2020A-Q13: 74 and 75).

⑦ Increment formula $\delta y \approx f'(a)\cdot\delta x$

- Write δx (⚠ 1° = π/180 rad; 1 month = 1/12 year) → differentiate at the starting point → multiply; if asked for the new value, add f(a).
- Example: $\ln 2.02 \approx \ln 2 + (1/2)(0.02) = 0.703$ (2021F-Q3).

⑧ Related rates / tangents

- $dy/dt = (dy/d\theta)(d\theta/dt)$: first use the geometric relationship to write the function (lighthouse $y = 12\tan\theta$) → differentiate → substitute the values at that time (first use y = 5 to find $\cos\theta = 12/13$).
- Tangent: the point of tangency satisfies $f'(a)$ = gradient, $f(a)$ = value on the tangent; tangent equation $y - f(a) = f'(a)(x - a)$.

⑨ Describing and interpreting (1–3 marks each year)

- Piecewise description: "For … the company makes a loss; profit increases to a maximum of $… at x = …; then decreases."
- The maximum rate of increase is where the graph has its greatest gradient, i.e. the point of inflection (2025A-Q9(c)).

2 Integration and its applications

① Antiderivatives and finding f from f′

- $\int e^{kx} = e^{kx}/k$; $\int \cos kx = \sin kx/k$; $\int (ax + b)^n = (ax + b)^{n+1}/[a(n + 1)]$ $(n \neq -1)$; $\int f'/f = \ln f(x) + c$ (in exams, usually $f > 0$).
- Make the numerator into the derivative of the denominator: $\int (3x + 1)/(3x^2 + 2x + 1) = \frac{1}{2}\ln(\ldots)$ (adjusting the coefficient is worth 1 mark separately).
- Find f from f′: integrate and write + c → substitute the known point → write the full f(x), simplifying with logarithm laws if needed.

② "Hence" integration (common in CF)

- Fixed three steps: integrate both sides → cancel the left-hand side using FTC → split the right-hand side and rearrange.

- Example: d/dx(x ln x) = ln x + 1 ⇒ ∫ln x dx = x ln x − x + c. This is often followed by a part asking for expected value, a definite integral or area.

③ Definite integrals from graphs

- Split the region into rectangles, triangles and sectors; above the x-axis is positive, below is negative. ∫(f − 2) = ∫f − 2 × interval length; $\int_a^b f' = f(b) - f(a)$ (read the two values directly from the graph).
- Find α such that ∫₀^α f = 0: set the area above equal to the area below.

④ FTC and functions defined by integrals ★ (flagged repeatedly in reports)

- $\frac{d}{dx}\int_a^x f(t)dt = f(x)$ (answer as a function of x); if the upper limit is g(x), multiply by g′(x), e.g. D(t) = ∫₀^(πt) … ⇒ D′(t) = π·(…).
- If the lower limit is the variable: $\int_t^3 f = -\int_3^t f$ ⇒ derivative is −f(t).
- Sketch A(x) = $\int_a^x f$: A(a) = 0; f changes from positive to negative ⇒ A has a maximum, from negative to positive ⇒ A has a minimum; a turning point of f ⇒ a point of inflection of A; label the value of A at endpoints.
- Determine the sign of F(x): compare the positive and negative areas between 0 and x, and refer to areas in your reasoning.

⑤ Area

- First find intersection points (often worth 1 mark separately) → ∫(top − bottom)dx → the integral expression must include limits and dx (2025F-Q5 gives 1 mark specifically for notation).
- Cases requiring splitting: finding area between the curve and the x-axis when the curve crosses the x-axis; or when the top/bottom relationship of two curves swaps. Also split when the upper/lower boundaries are made up of multiple functions (2024 report).
- Rectangle subtraction: area enclosed by $y = e^x$, y = 2 and the y-axis = 2 ln 2 − ∫₀^(ln2) e^x dx = 2 ln 2 − 1.
- Inverse function area: ln x lies below the x-axis when x < 1, so add a negative sign (2020A-Q11(e)).

⑥ Definite integral equations with parameters: evaluate the expression containing the parameter = given value, then solve (ln b − ln a = ln 3 ⇒ b = 3a).

⑦ Riemann sums

- For a lower sum take the smaller function value on each subinterval, for an upper sum take the larger; for increasing functions the left endpoint underestimates and the right endpoint overestimates; for decreasing functions it is the opposite (1/x). The explanation is worth 1 mark separately.
- Best estimate = (L + U)/2; ways to improve: add more rectangles, or find the function and integrate.

⑧ Contextual applications: cross-sectional area × length (width) = volume; total change = ∫rate of change; when the water level is unknown, first write the cross-sectional area as a function of h (with h in the upper limit), then solve for h; make sure units are consistent (cm and m).

3 Rectilinear motion

Concept comparison

- Displacement x can be positive or negative; "at rest" ⇔ v = 0; changing direction ⇔ v changes sign (v touching 0 and returning to the original sign does not count, e.g. v = (t − 2)²).
- Returning to the starting point ⇔ x(t) = x(0); passing through the origin ⇔ x(t) = 0 (the two are the same only when x(0) = 0).
- Time terminology: "first appears / initially" is t = 0 (2024 report: many substituted t = 1); "during the fifth second" is 4 ≤ t ≤ 5.

① Speeding up or slowing down ★

- v and a have the same sign ⇒ speeding up; opposite signs ⇒ slowing down. ⚠ Just saying "acceleration is negative" does not get the mark (exact wording in the 2024F-Q2 key).
- ⚠ The 2018A-Q11(c) key treats "heading south and decelerating at 0.5" as a = −0.5; for wording like this, state your interpretation and positive direction.

② Working backwards by integrating a(t) or v(t)

- Add a constant at each integration; use v(0), x(0) to determine the constants; with two position conditions, solve simultaneously to find the parameters and constants (2017A-Q20: p = 12).
- For trigonometric integrals, divide by the inner coefficient: ∫2 sin(t/3 + π/6) dt = −6 cos(t/3 + π/6) + C.

③ Distance travelled and change in displacement ★ examined every year

- Distance travelled: solve v = 0 to find all stopping times → split into intervals → put a negative sign in front of the integral for the interval where v < 0 (or add absolute values for each interval) → sum with units. The sign is worth 1 mark on its own, and writing dt in every integral is also worth 1 mark (2023A-Q8(c)).
- It can also be written as |x(t₁) − x(0)| + |x(t₂) − x(t₁)| + … (2024F-Q2(d)).
- Interpreting $\int_a^b$ v dt: "the change in displacement from t = a s to t = b s" (the start and end times must be written; 1 mark on its own).
- ⚠ Do not directly use x(final) − x(initial) as distance travelled.

④ Maximum velocity / acceleration zero

- At an interior extremum of the interval, a = 0 must occur, but the converse is not necessarily true (in a constant-velocity interval, a is always 0); verify using v″ < 0 or a sign table (marks are available when the question requires using calculus).
- When finding maximum speed, also compare |v| at the endpoints.

⑤ Interpreting graphs and sign tables: Use the sign of a to explain whether v is increasing or decreasing; the two parts before and after are worth 1 mark each; when there is only a sign table for x, v, and a: x changes sign ⇒ passes through the origin, v changes sign ⇒ stops and turns around, v and a have opposite signs ⇒ slowing down.

4 Exponentials and logarithms

① Logarithm laws and expressing in letters (CF)

- log(mn) = log m + log n; log(m/n) = log m − log n; log m^k = k log m; change of base $\log_a b = \log_c b / \log_c a$.
- Write constants as logarithms: 4 = ln e^4, 2 = $\log_{10}100$, 9 = $\log_{10}10^9$ (this is a key step in several questions).
- Example: p = ln 2, r = ln 5 ⇒ ln 6.25 = ln(5²/2²) = 2r − 2p; m = $\log_3 6$, n = $\log_6 5$ ⇒ $\log_3 10$ = (m − 1) + mn; e^(ln2 + ln3) = 6.

② Solving exponential / logarithmic equations (CF exact values)

- Collect like terms: $4e^{2x} = 81 - 5e^{2x} \Rightarrow e^{2x} = 9 \Rightarrow x = \ln 3$.

- Quadratic in e^x: let $u = e^x$, factorise, discard $u \le 0$.
- Logarithmic equations: first combine into a single logarithm, then remove the log; ⚠ check the domain, 2 ln x = … only the positive root is allowed (2025F-Q7 only awards marks for one solution).

③ Logarithmic graphs and transformations

- $y = m \log_a(x - p) + q$: vertical asymptote $x = p$, passes through $(p + 1, q)$; determine p from the asymptote, then substitute two points to solve for m and q.
- Graphing is worth 3 marks: asymptote (draw it and write its equation), one special point, correct shape.
- Transformations: ln 4x = ln x + ln 4 (vertical translation up); ln√x = ½ ln x (vertical dilation); ln x + 4 = ln(e^4x) (horizontal dilation); $\log_2(1/(x - 1)) = -\log_2(x - 1)$.
- When reading values from a graph, show the reading process on the graph (2024A-Q15 "show clearly how you used the graph").

④ Growth / decay model $P = P_0e^{kt}$ ★ high frequency in CA

- Initial value: t = 0; find k from the half-life ½ = e^(k·T½); reaching a certain value: set the model equal to the target value and solve using ln.
- Convert to a time and round as required by the question: e.g. 4e^(−0.05t) = 2.5 ⇒ t ≈ 9.40 h ⇒ 9 am + 9 h 24 min = 6:24 pm (for "latest", round down; 6:25 is already too late).
- Rate of change dP/dt = kP: if asked for rate of change, give the signed value; if asked for rate of decrease, write it as a positive number and state decreasing.
- Piecewise model: the initial value of the later piece equals the value at the end of the earlier piece; long-term behaviour: as t → ∞, e^(−kt) → 0.

⑤ Logarithmic scales (decibels, Richter, site rankings)

- $L = k\cdot\log(I/I_0) \Rightarrow I_2/I_1 = 10^{((L_2 - L_1)/k)}$. ⚠ k differs between questions: decibels k = 10 (90 dB is 1000 times 60 dB); sound pressure k = 20; Richter k = 1; site ranking k = 2.
- The difference in readings is not the answer; it must also be converted to a power of 10; to find I_0: substitute a set of (L, I) and then convert to exponential form.
- log P is a straight line against t ⇒ P = 10^B·(10^A)^t; use the slope to compare growth rates.

⑥ Logarithmic model + differentiation: determine parameters from known points (A(0) = 21, A(1) = 53 ⇒ c = 21, b = 64); ds/dD = 40.6/D decreases as D increases, use this to draw a conclusion; round quantity down; 1 < x < 2 belongs to "week 2".

5 Discrete random variables and the binomial distribution

① Distribution tables

- For equally likely outcomes, list the sample space (for two dice, use a table); when there are unknowns, use ΣP = 1 and E(X) = μ to set up two equations (1 mark each).
- From the cumulative distribution P(X ≤ x): P(X = k) = F(k) − F(k − 1); conditional probability P(X = 1 | X ≤ 3) = P(X = 1)/P(X ≤ 3).
- For a new variable, split into cases: number unsold Y = max(0, 3 − X).

② Expectation, variance, linear transformations

- $E(X) = \Sigma xp$; $Var(X) = \Sigma x^2p - \mu^2$; E(aX + b) = aE(X) + b; Var(aX + b) = a^2Var(X); SD(aX + b) = |a|·SD.
- ⚠ Var(25 − 2X) = 4Var(X); a constant does not affect the variance.
- Scaling to a new mean and standard deviation: find a from the SD first (take a > 0 to preserve the ranking order), then find b from the mean. A cost increase of 20% plus an extra $1 ⇒ new mean 1.2μ + 1, new SD 1.2σ.

③ Game / fundraising expectation (examined in 6 out of 10 years)

- First define the payoff variable; be clear whose perspective you are taking (player or organiser) → make a "net payoff + probability" table → E = Σ(value × probability) → state a conclusion ("on average / in the long run").
- Pricing: set the expected profit equal to the target; minimum charge = expected payout (2025A-Q8: E = 0.8, $5 per red ball ⇒ at least $4).

④ Binomial distribution ★

- Write X ~ Bin(n, p); usually the name is worth 1 mark and the parameters 1 mark; state what n and p mean in context.
- E = np, Var = np(1 − p); work backwards from the mean and variance: np = ½, np(1 − p) = 5/12 ⇒ p = 1/6, n = 3.
- Boundaries: more than three ⇒ X ≥ 4; system has fewer than 2 working ⇔ at least 4 broken; change parameters when the situation changes (remove one ⇒ Bin(4, p)).
- Assumptions (2 marks): trials are mutually independent; the probability of success p is the same and constant, write in context. "Two outcomes" and "n fixed" are usually given in the question; writing them may not earn marks.
- Not binomial: the number of trials is not fixed; sampling without replacement makes the trials not independent (improvement: sample with replacement).
- For very small probabilities, keep more decimal places (2019A-Q18(c) about 0.00003; the key requires at least 5 decimal places).

⑤ Normal → binomial: first use the normal distribution to find the single-item probability p (keep enough decimal places) → Y ~ Bin(n, p) (write "binomial" and the parameters) → find P(Y ≥ k).

⑥ Paths and counting: translate the event into the number of successes. Example: returning to the origin ⇔ number of forward steps = number of backward steps (impossible with an odd number of steps, probability 0; write a reason); moving forward ≥ 20 cm (5 cm per step) ⇔ at least 7 forward steps out of 10.

6 Continuous random variables and the normal distribution

① Histogram → probability (CF)

- Add relative frequencies; for a frequency histogram, first divide by the total number.
- If an endpoint falls inside a bar, interpolate proportionally (provided the question states "uniform distribution within classes"); E ≈ Σ class midpoint × probability; distinguish cumulative probability tables from individual probabilities.

② The four pdf essentials

- Total area = 1 (write this step when finding k or proving k); $P(a < X < b) = \int_a^b f$; $E = \int xf$; $Var = \int x^2f - \mu^2$ or $\int(x - \mu)^2f$; median ∫^M f = 0.5.
- Given cdf F(x): P(a < X ≤ b) = F(b) − F(a); pdf = F′(x).

- In CF papers, for pdfs made of straight-line segments, use triangle and trapezium areas; for piecewise pdfs, integrate each piece separately.
- Example: $f = 1/(x \ln 5)$, $1 \le x \le 5 \Rightarrow E = 4/\ln 5$, median $\sqrt{5}$.
- Convert the expectation to the context units or time (3.8 h after noon ⇒ 3:48 pm); the units of variance are squared units.

③ Uniform distribution

- $f = 1/(b - a)$, write the domain $a \le x \le b$ and 0 otherwise; $E = (a + b)/2$.
- ⚠ For variance, answer with an integral expression (the key gives 1 mark each for "pdf and domain" and "integral expression"); $(b - a)^2/12$ is not on the formula sheet, so use it only to check. Example: $E = 6$, maximum value 9 ⇒ $U(3, 9)$, $\text{Var} = \int_3^9 (x - 6)^2/6 \, dx = 3$.
- Conditional probability: $P(X < -0.35 \mid X < 0) = 0.15/0.5$; $X^2 < 0.09 \Leftrightarrow -0.3 < X < 0.3$.

④ Normal distribution

- Write the probability statement first (most keys award 1 mark separately) then calculate; for inverse quantile: $P(X > m) = 0.0005$ ⇒ invNorm(0.9995).
- Conditional probability: $P(X < 75 \mid X > 67) = P(67 < X < 75)/P(X > 67)$.
- Linear transformation $Y = aX + b$: $\mu_Y = a\mu + b$, $\sigma_Y = |a|\sigma$, $\text{Var}(Y) = a^2\text{Var}(X)$; you may also have to work backwards to find a, b from μ_Y, σ_Y.

⑤ Finding μ, σ from two tail probabilities ★

- Find two z-values (1 mark) → set up simultaneous equations → solve. Example: 5% below 55, 13% above 100 ⇒ $z = -1.6449$, 1.1264 ⇒ $\sigma \approx 16.24$, $\mu \approx 81.71$.
- $P(X \le 150) = 0.0228$, $P(X \ge 165) = 0.1587$ correspond exactly to $z = -2$, $z = 1$ ⇒ $\sigma = 5$, $\mu = 160$. When only the mean is known, one z is enough.

⑥ 68–95–99.7 (CF): write values as $\mu \pm k\sigma$; the flatter and wider the curve, the larger σ; the probability of an interval of width σ is at most about 0.38.

⑦ Reasons a normal model is unsuitable: histogram is asymmetric or skewed; variable is bounded; model gives impossible values (e.g. $P(\text{weight} < 0) > 0$).

7 Sample proportions and confidence intervals

① Distribution of $\hat{p}$

- Write "approximately normal (n sufficiently large) + mean p + variance $p(1 - p)/n$", 3 marks in total, then find the probability.
- Distribution is not symmetric (e.g. $n = 36$, $p = 0.05$) ⇒ the normal approximation cannot be used.
- As n increases ⇒ the standard deviation of $\hat{p}$ decreases and the distribution becomes more concentrated ⇒ probabilities such as $P(\hat{p} \le 0.03)$ increase (1 mark for each point).

② Calculating a confidence interval

- $\hat{p} \pm z\sqrt{\hat{p}(1 - \hat{p})/n}$, $z = 1.645$ (90%) / 1.960 (95%) / 2.576 (99%); round to the required number of decimal places.
- Work backwards from the interval: $\hat{p}$ = midpoint, E = half-width; given the interval and n, find the confidence level: $z = E/\text{SE}$, then look up the percentage.

③ Sample size ⚠ most error-prone

- When finding the minimum sample size (minimum / guarantee / ensure margin at most), always use $\hat{p} = 0.5$, even if $\hat{p}$ has already been calculated earlier (2023A-Q7(c): using 0.5 gives 2401, substituting 0.38 gives 2263, losing marks). Only substitute a particular sample proportion if the question explicitly says to use it.
- "At least" ⇒ round up (259.2 ⇒ 260); working back from an interval to the original n ⇒ round to nearest integer (350.1 ⇒ 350).
- Proportional relationships: halving the error ⇒ n × 4; width becomes 1/3 ⇒ n × 9.

④ Using an interval to test a claim ★ appears every year

- Two sentences: "The claimed value … is (not) within the CI …" + "so there is sufficient / insufficient evidence to conclude …". ⚠ You must take a position; you cannot avoid it by saying "cannot be 100% certain" (2023 report).
- Two intervals overlap ⇒ insufficient evidence to say the two are different.
- If asked "does it prove … / was a mistake made" ⇒ answer No: not all intervals contain p, and we do not know whether this interval happened to miss it (2025 report).

⑤ Factors affecting interval width (1 mark for factor + 1 mark for effect): increasing n makes it narrower; increasing the confidence level makes it wider; the closer $\hat{p}$ is to 0.5, the wider it is.

⑥ Sampling bias

- How to write: identify the source + explain which types of people are more or less likely to be sampled + link to the context; time and location are two sources and must be written separately (2022 report).
- Common sources: location (only outside shopping centres and libraries), time (weekday lunchtime), voluntary response, leading questions, sampling frame (mobile phones only), convenience sampling (the first 200 copies printed).
- Improvement: random sampling across the whole population (the whole week, all printing presses).

⑦ Other common question types

- The number of p-containing intervals among k 90% intervals ~ Bin(k, 0.9); "3 intervals do not contain p" ⇔ $X = k - 3$.
- Assumptions of the binomial distribution: write the assumptions → decide whether they are reasonable in context → give reasons.